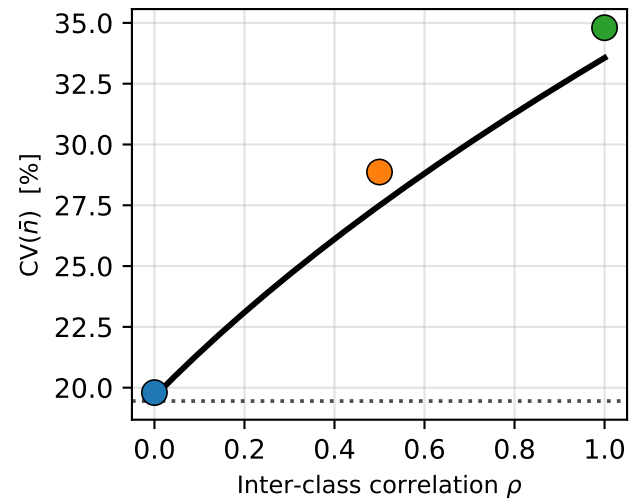


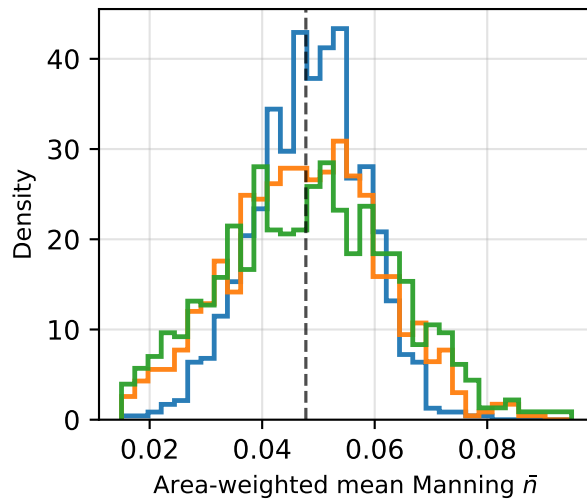
Effect of inter-class Manning correlation on spatial-mean roughness and flood outputs

(a) CV amplification of $\bar{n}$



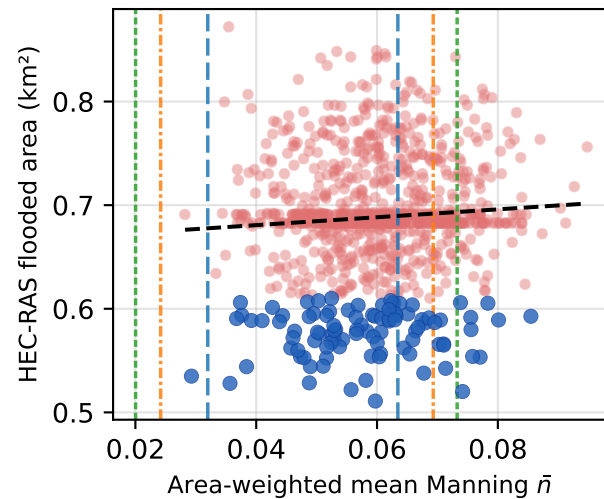
— Theory
... Current MC (19.4%)
 $\rho = 0$ (19.8%)
 $\rho = 0.5$ (28.9%)
 $\rho = 1$ (34.8%)

(b) Distribution of $\bar{n}$ by scenario



$\rho = 0$
 $\rho = 0.5$
 $\rho = 1$
-- Mean $\bar{n}$

(c) $\bar{n}$ vs flood output (995 runs)



High regime ($N = 898$)
Low regime ($N = 97$)
All: $r = 0.071$, $p = 0.03$
 $\rho = 0$ range
 $\rho = 0.5$ range
 $\rho = 1$ range